

EDUCATION	Arizona State University	Tempe, AZ
	<i>Ph.D. Student in Computer Science</i>	2023 - present
	• Research area: Software Engineering, Program Analysis, Machine Learning • <i>Advisor: Prof. Xusheng Xiao</i>	
	University of Washington	Seattle, WA
	<i>Master of Science in Electrical & Computer Engineering</i>	2019 - 2021
	Cuiying Honors College, Lanzhou University	Lanzhou, China
	<i>Bachelor of Science in Physics, with highest Honors</i>	2015 - 2019
PUBLICATIONS	1. Shang Ma, Chen Yanai, Avichai Ben, Zichen Liu , Yanfang Ye, Xusheng Xiao. “Demystifying Scam Playbooks: Psychological Techniques, Scenario Taxonomy, and Conversation-Aware Detection.” <i>Under submission to the IEEE Symposium on Security and Privacy (S&P)</i> , 2027. [arXiv]	
	2. Zichen Liu and Xusheng Xiao. “AppBDS: LLM-Powered Description Synthesis for Sensitive Behaviors in Mobile Apps.” <i>Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE)</i> , 2025. [Paper]	
	3. Zichen Liu , Shao Yang, Xusheng Xiao. “BINCTX: Multi-Modal Representation Learning for Robust Android App Behavior Detection.” <i>arXiv preprint arXiv:2510.14344</i> , 2025. [arXiv]	
	4. Jiong Xiao Wang*, Zichen Liu* , Keun Hee Park, Muhao Chen, and Chaowei Xiao. “Adversarial Demonstration Attacks on Large Language Models.” <i>arXiv preprint arXiv:2305.14950</i> , 2023. [arXiv]	
	5. Cheng-Yen Yang*, Aotian Zheng*, Zichen Liu , and Jenq-Neng Hwang. “Tracklet-based Multi-camera Multiple People Tracking.” In <i>International Conference on Computer Vision (ICCV) Workshops</i> , 2021. Ranked 3rd out of ~80 competitors. [Presentation]	
RESEARCH EXPERIENCE	Research Assistant RISE Lab @ASU	
	<i>Advisor: Prof. Xusheng Xiao</i>	2023.7 - present
	<i>Understanding and Disrupting the Online Scam Ecosystem</i>	
	Overview: Scammers reuse a small number of playbooks, embedding the same psychological techniques in recurring scenarios. I study that structure at scale and turn it into defenses that act before the money moves. • EARS, conversation-aware scam detection: built a detector that infers which scam scenario a customer is describing from a partially told story, pretraining scenario semantics on large-scale incident reports and modeling the phase of each conversational turn so that the model stays silent until diagnostic evidence appears, then raises a warning while the call is still in progress • Longitudinal measurement of the scam ecosystem [Ongoing]: built an eight-year, 611K-report corpus end to end, and am tracing how scam scenarios, campaign organization, contact channels and persuasion effectiveness co-evolve	
	<i>Structuring Security Knowledge for LLM-Guided Vulnerability Repair [Ongoing]</i> Overview: Retrieval-guided LLM repair treats two patches as interchangeable when their edits look alike. I am building the knowledge that distinction needs: what a vulnerability violates, and what its patch does. • Working toward conditioning LLM repair on those two paired axes. So far I have distilled candidate fix-pattern families by clustering the edits of real patches, and built a mechanism-level security-impact taxonomy that names the concrete safety constraint each vulnerability violates	

Accountable Mobile Software: Explaining Sensitive App Behavior at Scale

Overview: Mobile apps request dangerous permissions far more often than they explain what they do with the data. I recover that explanation from the app itself, with static analysis plus LLM reasoning.

- **AppBDS:** introduced the UI-Fused Call Graph, which anchors static analysis to the interface a user actually sees so that the program surface collapses to its privacy-relevant fraction
- Paired it with a multi-agent LLM pipeline that cross-validates evidence before describing how an app uses each sensitive permission, and showed the result holds under code obfuscation and across both open and proprietary LLM backends

Research Assistant | SaFo Lab @ASU

Advisor: Prof. Chaowei Xiao

2023.2 - 2023.7

Adversarial In-Context Learning: Attack Surface and Stealth

Overview: In-context learning lets an LLM be reprogrammed by the examples in its prompt, which makes those examples a target. I studied whether tampering with only the demonstrations can steer the model unnoticed.

- Pioneered a threat model in which perturbing only the in-context demonstrations, never the user's query, steers an LLM to targeted misclassification, establishing in-context learning itself as an attack surface
- Showed the perturbations survive linguistic and semantic inspection, perplexity screening and human review, so the attack cannot be caught by inspection alone

Research Assistant | Information Processing Lab @UW

Advisor: Prof. Jenq-Neng Huang

2020.12 - 2021.9

Cross-View Human Perception: Multi-Camera Tracking and Pose Estimation

- Built a multi-camera tracking system whose cross-camera tracklet association lets a single-camera baseline generalize across views, and proposed **Joint-Flow**, fusing top-down, bottom-up and temporal cues for pose tracking

Machine Learning Engineer | Ernst Leitz Labs, Leica Camera AG 2021.09 - 2022.11

- Developed deep-learning replacements for stages of the mobile camera ISP pipeline, and built the rendering engine and depth estimation behind the Leitz Looks Engine
- Ported depth models such as RAFT-Stereo and MiDaS onto a next-generation Qualcomm Snapdragon chipset, with speed and power optimization and performance profiling

INDUSTRY EXPERIENCE

TEACHING EXPERIENCE

Teaching Assistant | Arizona State University

- CSE 464 Software Quality Assurance and Testing Fall 2023 – Fall 2025
- CSE 471 Introduction to Artificial Intelligence, Spring 24

ACADEMIC SERVICE

- **Artifact Evaluation Committee:** IEEE Symposium on Security and Privacy (S&P) 2027, ISSTA 2026, ASE 2026
- **Reviewer:** NeurIPS 2026 FLLMPT Workshop

SKILLS

Languages: Python, Java, C/C++

ML & LLM Tooling: PyTorch, Hugging Face Transformers, PEFT, vLLM

LLM Methods: parameter-efficient fine-tuning, few-shot and retrieval-based prompting, multi-agent LLM pipelines, LLM-assisted annotation with active learning

Program Analysis & Security: Soot/SootUp, FlowDroid, Android/APK Analysis, Call-Graph & Taint Analysis

Tools & DevOps: Git, Docker, CI/CD

PROJECTS

Blockchain Consensus Simulator

2020.10 - 2020.12

Adversarial Attacks and Defenses: A Survey

2020.5 - 2020.6